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EDUCATION

Babu Banarasi Das Institute of Technology and Management

B.Tech in Computer Science and Engineering

Lucknow, India

2024 – 2028

PROJECTS

APIx: Daily Airfare Price Index for India | *Python, PostgreSQL, Next.js, GitHub Actions* Live | GitHub

- Built a daily collection pipeline on GitHub Actions that prices 12 DGCA-weighted routes across 5 advance-purchase windows, storing **24,448 fares** in Postgres
- Implemented the official CPI method in Python (Jevons elementary indices, Young aggregation, outlier cleaning, imputation), **reproducing MoSPI's published worked examples to 4 decimal places**
- Made every number auditable: each index point carries a hash of its inputs, and each publish writes a full new version in one transaction so readers never see half-written data
- Served the index via 19 public JSON endpoints with a static-snapshot fallback; **79 pytest tests**

DataPilot: Prompt-to-Dataset LLM Agent | *TypeScript, Next.js, PostgreSQL, LLM APIs* Live | GitHub

- Built an agent pipeline where an LLM planner turns a plain-English request into a schema and plan, runs **9 source connectors in parallel**, and extracts rows in batched LLM calls
- Guarded against hallucinated rows by requiring a verbatim source quote per row, **verified in code**, then validating types, merging duplicates and scoring each row's confidence
- Hardened it against untrusted web input with an SSRF guard (private, loopback and cloud-metadata IPs) and by fencing scraped text as data to resist prompt injection
- Wrote a provider-agnostic LLM client (Gemini, Anthropic, OpenAI) with model fallbacks and a rule-based mode, plus an eval harness scoring fill rate and precision

Curio: Personalized Reading Feed | *TypeScript, Next.js, PostgreSQL, Drizzle ORM, Resend* Live | GitHub

- Ingested **275+ sources across 26 topics** (RSS, Hacker News, Reddit, YouTube) in parallel with an 8-second per-source timeout, so one slow feed never stalls a run
- Ranked feeds by blending recency with a per-source quality score from bookmark and read rates; built search on Postgres full-text search instead of an external service
- Automated 5 scheduled jobs (ingestion, digests, quality scoring, cleanup, re-engagement) and sent timezone-aware digests with HMAC-signed unsubscribe links; **80+ Vitest tests**

OPEN SOURCE

corsairdev/corsair | *13k+ stars, TypeScript* PR #703, merged

- Added the OCR.space integration: typed client, error handling and Jest tests (2,300+ lines, 17 files)

bholmesdev/hubble.md | *1.4k+ stars, Markdown editor* PR #241, merged

- Fixed the editor's Enter handler dropping users out of a list when pressing Enter after an image in a list item

twitbruv/twitbruv | *TypeScript* PR #113, merged

- Handled @-prefixed queries in user search and added a fallback for avatars that fail to load

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, SQL

Frameworks: Next.js, React, Node.js

Databases: PostgreSQL, Drizzle ORM, SQLite

AI: LLM APIs (OpenAI, Anthropic, Gemini), structured JSON output, LLM evals

Tools: Git, GitHub Actions, Vercel, Vitest, pytest

LEADERSHIP

Google Developer Groups (GDG) on Campus

Campus Lead

2026 – Present

BBDITM, Lucknow